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ROLLNO-CSB24068

```
#include<stdio.h>
```

```
#include<time.h>
```

```
#include<stdlib.h>
```

```
int linear(int a[],int n,int x)
```

```
{
```

```
    int i;
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
        if(a[i]==x)
```

```
        {
```

```
            printf("found");
```

```
            return 1;
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

```
int selection_sort(int a[],int n)
```

```
{
```

```
    int i,j,min;
```

```
    for(i=0;i<n-1;i++)
```

```

{
    min = i;
    for(j=i+1;j<n;j++)
    {
        if(a[j]<a[min])
        {
            min = j;
        }
    }
    int temp = a[i];
    a[i] = a[min];
    a[min] = temp;

}
return 1;
}

```

// constant time complexity

```

int getfirst(int a[])
{
    return a[0];
}

```

```

int main()
{

```

```

    clock_t start,end;

```

```
int n,i,m;

double c_time;

printf("Input\nEnter size of array : ");

scanf("%d",&n);

int a[n];

printf("here automatically stores values in array .");

for(i = 0;i<n;i++)
{
    a[i]=n-i;
}

int x;

printf("\nEnter value of x to be search : ");

scanf("%d",&x);

/*linear search performed */

start = clock();

linear(a,n,x);

end = clock();

c_time = ((double)(end-start))/CLOCKS_PER_SEC;

printf("Output\nTime taken by linear search O(n): %lf\n",c_time);

/* array gets sorted */
```

```
start = clock();
selection_sort(a,n);
end = clock();

c_time = ((double)(end-start))/CLOCKS_PER_SEC;
printf("Time taken by selection sort O(n^2): %lf\n",c_time);

/* Array is now sorted, 1st element get returned */
start = clock();
m=getfirst(a);
end = clock();

c_time = ((double)(end-start))/CLOCKS_PER_SEC;
printf("Time taken by constant O(1) : %lf\n\n",c_time);

return 0;
}
```

Output

Input

Enter size of array : 5

here automatically stores values in array .

Enter value of x to be search : 3

foundOutput

Time taken by linear search $O(n)$: 0.000004

Time taken by selection sort $O(n^2)$: 0.000001

Time taken by constant $O(1)$: 0.000001

=== Code Execution Successful ===